



TEACHING NOTE

Google's Project Oxygen: Do Managers Matter?

Synopsis

Google's Project Oxygen was launched in late 2009 with the goal of answering a question raised repeatedly by the company's engineers and executives: do managers matter? This question generated a multi-year research project that ultimately led to a comprehensive program, built around eight key management attributes, designed to help Google's managers become more effective. The case describes the design and execution of the research project, the rollout of the project's findings to employees, and the process of institutionalizing the desired behaviors through surveys, feedback, coaching, and training programs. It also provides representative examples and best practices associated with each of the eight management attributes.

By November 2012, the eight attributes had been widely adopted within Google, and the company was able to show statistically significant improvements in several important measures of managerial effectiveness and performance. Now, Prasad Setty, vice president of people analytics and compensation, must decide how his team can best build on the success of Project Oxygen. He is considering several possible initiatives, each aimed at understanding and improving another aspect of management at Google: the effectiveness of directors, vice presidents, and other senior executives; the composition and functioning of teams; the employee life cycle, with special attention to hiring, promotion, and lateral transfers; and the possibility of creating a larger number of "truly amazing managers."

Purpose

The case has three main purposes. First, it shows how companies can use data-driven, evidence-based approaches to improve the practice of management. Often called "competing through analytics" or "big data," these approaches involve the collection, compilation, and analysis of large amounts of data to draw inferences about customers, product and service offerings, and employees.¹

¹ For early discussions of these approaches, see Thomas H. Davenport, "Competing on Analytics," *Harvard Business Review*, January 2006, pp. 99-107, and Thomas H. Davenport and Jeanne G. Harris, *Competing on Analytics* (Boston: Harvard Business School Press, 2007). For more recent discussions, see Thomas H. Davenport, Paul Barth, and Randy Bean, "How 'Big Data' Is Different," *MIT Sloan Management Review*, Fall 2012, pp. 43-46; Andrew McAfee and Erik Brynjolfsson, "Big Data: The

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Initially, these approaches were used largely in product development, marketing, and pricing decisions; more recently, they have been applied to human resource management.² The case presents students with a rigorously executed example that incorporates many of the best practices associated with these techniques.

Second, the case provides insight and evidence about what constitutes effective management. Especially when compared with leadership, management has received comparatively limited attention from scholars in recent years. It remains surprisingly understudied and undertaught.³ In part, this reflects the difficulty of describing, precisely and concretely, what managers actually do. It is often said that “managers get things done through other people,” yet how they do so is seldom spelled out in detail.⁴ Project Oxygen, in contrast, is highly granular in its conclusions; it pinpoints specific behaviors and practices that are associated with increased management effectiveness. These behaviors provide a framework and set of categories for students to discuss their own experiences with good and bad managers and to assess their own strengths and weaknesses as (potential) managers.

Third, the case helps students understand the steps in a large-scale organizational change process. The goal of Project Oxygen was to introduce new ways of behaving and interacting at Google by transforming the culture from one in which management was viewed as an impediment to one in which management was viewed as valuable and improvements in management practice were actively pursued. In that sense, Project Oxygen can be viewed as a planned change process, driven by members of the people analytics group, designed to shift attitudes, behaviors, and ways of working at Google. Like most change initiatives, it proceeded through three broad stages: unfreezing, changing, and refreezing.⁵ Students explore the key elements of each of these stages by examining the research, rollout, and institutionalization of Project Oxygen; in the process, they develop a number of lessons about how to lead change successfully.

The case was originally designed for a general management course focused on implementation, but it can also be used in courses on leadership, organizational behavior, human resource management, and business policy. It is also appropriate for more specialized courses on analytics, big data, and managing high-technology firms.

Management Revolution,” *Harvard Business Review*, October 2012, pp. 60-68; and Steve Lohr, “Sizing Up Big Data,” *New York Times*, June 20, 2013, pp. F1, F10.

² Thomas H. Davenport, Jeanne Harris, and Jeremy Shapiro, “Competing on Talent Analytics,” *Harvard Business Review*, October 2010, pp. 52-58, and Steve Lohr, “Big Data, Trying to Build Better Workers,” *New York Times*, April 21, 2013, Sunday Business, p. 4.

³ Leonard R. Sayles, “Whatever Happened to Management...or Why the Dull Stepchild?” *Business Horizons*, April 1970, pp. 25-34; Henry Mintzberg, *Managers Not MBAs* (San Francisco: Berrett-Koehler, 2004), pp. 33-36; and Henry Mintzberg, *Managing* (San Francisco: Berrett-Koehler, 2011). In the latter book, Mintzberg is blunt in his assessment (p. 9): “I believe that we are now overled and undermanaged.”

⁴ The quotation is attributed to Mary Parker Follett. See Mintzberg, *Managing*, p. 12.

⁵ This is the classical model of change first proposed by Kurt Lewin. See Kurt Lewin, *Field Theory in Social Science* (New York: Harper and Brothers, 1951). A number of scholars have developed other, closely related three-stage models – for example, present state, transition state, future state and awakening, envisioning, enabling. For a discussion of these models, see Rosabeth Kanter, Barry A. Stein, and Todd D. Jick, *The Challenge of Organizational Change* (New York: Free Press, 1992), pp. 375-377. There are several more refined seven- or eight-stage models of change, but they are little different in substance from the three-stage models.

Suggested Assignment Questions

1. How would you describe Google’s culture? How were managers viewed before Project Oxygen?
2. What kinds of people work at Google? What should be the role of HR in such a setting? What should be the role of managers?
3. What is your evaluation of the steps that the PiLab took to:
 - Identify the characteristics of effective managers at Google?
 - Roll out Project Oxygen to the organization?
 - Ensure that the Oxygen 8 attributes were widely adopted and practiced?
4. What is your assessment of the Oxygen 8 attributes? What value (if any) do they provide? How generalizable are they?
5. What should Setty’s priorities be going forward? Which of the proposed initiatives should he pursue? In particular, should he and his team take on the challenge of trying to create “truly amazing managers?” If so, how?

Analysis

Culture and People

Google’s culture has several defining features.⁶ It is engineering-dominated (“a company built by engineers for engineers”); collegial, consensus-oriented, and non-hierarchical (like a “college campus”); and family-like and developmentally-focused (executives have “a relationship with our employees” and strive to “figure out the best things we can do for them”).

Google was founded by and continues to be dominated by engineers, who have a distinctive ethos and culture.⁷ Engineers value individual autonomy and are strongly committed to getting the work done, often through long hours and devotion to the task. They see authority as stemming primarily from technical competence, knowledge, and expertise rather than position or rank. They are committed to data-driven decision making and seek to find the “right answer,” which they only accept if it meets rigorous standards of proof. They attach great importance to technology and thoughtful design, at times elevating them over personal relationships. Not surprisingly, these characteristics lead many engineers at Google (and elsewhere) to question the value, importance, and

⁶ Scholars have offered many different definitions of organizational culture. Some have focused on shared philosophy and assumptions, others have focused on norms and espoused values, and still others have focused on observed regularities and patterns of thinking and behaving. See Edgar H. Schein, *Organizational Culture and Leadership*, Fourth Edition (San Francisco: Jossey-Bass, 2010), pp. 14-18 for a list of many of these factors, as well as an inclusive definition.

⁷ Strictly speaking, Google is dominated by software programmers and computer science graduates, not engineers. However, according to an expert on organizational culture, these groups have similar values: “...engineers includes the technocrats and core designers in any functional group. For example, the designers of information technology systems might be software programmers, but I would include them in the concept of the engineering culture.” See Edgar H. Schein, “Culture: The Missing Concept in Organizational Studies,” *Administrative Science Quarterly*, volume 4 (1996), p. 237.

contribution of managers, whom they view as taking time away from the task at hand, interfering with real work, and creating barriers to independence and design autonomy.⁸

In keeping with this ethos, Google has long been highly collegial. Many employees elevate peer respect over formal authority; they seldom accept top-down directives without question simply because of their source. The organizational structure is relatively flat, with few levels, reflecting a general distaste for hierarchy. Together, these characteristics lead to a preference for consensus-based decision making, with individuals more likely to back a decision or directive because they have reasoned their way to it rather than being told to comply by higher-ups. The existence of relatively few hierarchical levels also means that promotion, in the traditional sense of being given a job with a larger title, more direct reports, or higher rank, occurs relatively seldom at Google and often cannot be used as a motivator.

Google’s culture is also family-like, with a strong developmental orientation.⁹ Senior executives are committed to providing employees with opportunities for growth and personal development; in that sense, managers play an almost parental role. As Setty observes on page 11 of the case, “Treating people like family means nurturing them, developing them, and making sure they reach their full potential.” This includes helping employees with their careers and ensuring that they expand their repertoire of skills.

The company’s work force is highly talented, largely because Google is considered such a desirable employer and the selection process is so demanding.¹⁰ Most Googlers are smart, driven, high achievers, with a history of success. The company is, to an unusual degree, made up almost entirely of “A” players.¹¹ Employees are also young, hardworking, and ambitious; few are likely to have failed very often or received much in the way of critical, negative feedback. They are simultaneously collaborative and fiercely individualistic; they enjoy working together yet “crave autonomy.” They are flexible and adaptable, reflecting both temperament and necessity: the underlying technology is continuously in flux, the organization structure is fluid, and roles and assignments change rapidly and repeatedly. Many employees have relatively short tenures at the company, reflecting Google’s extraordinary growth rates and the constant influx of new people (headcount rose from 300 in 2001 to over 37,500 in 2012).

With this mix of culture and people, Google’s managers and HR function have demanding roles to play. Managers must be unusually attentive to, and supportive of, employees’ personal needs, development, and career planning. They must provide a steady stream of feedback to direct reports and team members, guiding them to superior performance and greater achievement. At the same time, they must be careful to intervene relatively infrequently in the work process; even then, they need to do so with a light touch, since too much hands-on direction and control is likely to be deemed “micromanaging.” They must be carriers and conveyers of the culture, introducing new,

⁸ Schein, “Culture: The Missing Concept in Organizational Studies,” Schein, *Organizational Culture and Leadership*, pp. 60-62; J. Gregg Robinson and Judith S. McIlwee, “Men, Women, and the Culture of Engineering,” *The Sociological Quarterly*, volume 32 (1991), pp. 403-421; and Paul M. Leonardi, “The Mythos of Engineering Culture: A Study of Communication Performances and Interpretation,” M.A. thesis, University of Colorado, Department of Communication, 2003.

⁹ For a concise description of the developmental perspective, including contrasts with the “Darwinian” approach taken by many firms, see Morgan W. McCall, Jr., *High Flyers* (Boston: Harvard Business School Press, 1998), pp. 11-16.

¹⁰ James B. Stewart, “A Place to Play for Google Staff,” *New York Times*, March 16, 2013, pp. B1, B4, and Boris Groysberg, David A. Thomas, and Alison Berkley Wagonfeld, “Keeping Google ‘Googley,’” Harvard Business School case 9-409-039, July 2, 2011, pp. 5-6.

¹¹ Groysberg, Thomas, and Wagonfeld, “Keeping Google ‘Googley,’” p. 5.

inexperienced employees relatively quickly to “the way things are done around here.” Perhaps most difficult, they must negotiate a delicate balancing act, keeping employees happy and motivated through enthusiastic cheerleading and support while at the same time helping them develop and grow through critical feedback and challenging “stretch” assignments. This balancing act is likely to be especially difficult because of the tension between what employees *want* (their desire to be left alone and largely unsupervised so that they can do the job their way) and what employees *need* (regular oversight, guidance, and critiques that will help them learn to do things better or differently, improving their performance while expanding their skills and capabilities).

In this setting, Google’s HR specialists must combine traditional and novel roles. They must ensure a continued inflow of exceptionally high-level talent, individuals who are simultaneously smart, accomplished, and likely to be a cultural fit. They must provide managers with processes and mechanisms for providing regular, personalized feedback to employees, especially those who are relatively new to the company. They must assist new managers in adapting quickly to their unfamiliar roles and must ensure that poor or ineffective managers are identified early and given the tools to improve.¹² Perhaps most important, they must play an enabling, facilitating role in ensuring employees’ satisfaction and well-being, making a convincing case for new policies and proposals using data and rigorous analysis to overcome engineers’ natural skepticism about the “soft stuff.”¹³

Project Oxygen as a Successful Change Process

Project Oxygen can be viewed as a highly successful planned change process, aimed at altering employees’ attitudes and behaviors. It was initiated and carried out by Google’s HR function (known internally as people operations). In a setting where managers were generally viewed unfavorably – to the point where Page and Brin, the co-founders, had for a brief period eliminated all managers from the engineering organization – members of the PiLab, a small team of statisticians and researchers within Google’s people analytics group, sought to gain greater acceptance for the importance of, and need to improve, management across the organization. They did so through a change process that unfolded in three broad stages: identifying the characteristics and impact of effective managers, rolling out the findings to the organization, and improving management effectiveness through widespread adoption and institutionalization of the Oxygen 8 attributes.

Identifying the characteristics and impact of effective managers Rigorous research was a necessary first step because of engineers’ perceptions and prevailing views about managers. The team’s goal at this stage was to anticipate likely objections and push back from employees and provide compelling evidence to overcome them. In particular, any data and supporting analysis had to meet engineers’ demanding standards of proof. Otherwise, proposed changes were unlikely to be accepted.

To make the case, members of the PiLab used only internal company data, clearly signaling that “the findings are about us.” They began by trying to prove the opposite case – that managers *don’t*

¹² The transition into a management role is difficult under most circumstances. According to a leading study: “In spite of their established qualifications as individual contributors, many new managers never adjust successfully to managerial responsibilities. First-line management is the level in the organization from which come the most frequent reports of incompetence, burnout, and excessive attrition. The human and financial costs are staggering...” See Linda A. Hill, *Becoming a Manager* (Boston: Harvard Business School Press, 1992), p. 2.

¹³ Farhad Manjoo, “The Happiness Machine: How Google Became Such a Great Place to Work,” http://www.slate.com/articles/technology/technology/2013/01/google_people_operations_the_secrets_of_the_world_s_most_scientific_human.html, accessed June 26, 2013.

matter – signaling that the team was bending over backward to be objective and neutral.¹⁴ They identified best and worst managers by combining the evaluations of superiors and subordinates, increasing the likelihood that the right people were in the sample.¹⁵ They compared upper quartile and lower quartile managers using refined statistical techniques that established causal links between better management and lower turnover, increased employee satisfaction, and other performance measures. Focusing on causality was essential because, in the words of one member of the team, “Google has high standards of proof...Simple correlations weren’t going to be enough.” Once the links between management effectiveness and key performance measures were firmly established, they launched a series of qualitative, double-blind interviews to identify the key attributes and behaviors associated with management effectiveness.

At this stage of the change process, the PiLab team had to meet several requirements for success. All involved overcoming the deeply ingrained skepticism of the company’s engineers about the value-added of managers. To that end, the team had to conduct their research in as an objective, scientific manner as possible to ensure that the findings were viewed as credible and unimpeachable. They had to show meaningful performance differences attributable to better management to motivate changes in behavior. And they had to develop sufficiently concrete, granular guidelines for practice that managers could not claim that the path forward was fuzzy and unclear. The quality of the research – its careful design, disciplined execution, and strength and detail of the findings – was the foundation for moving forward with a successful change initiative.

Rolling out the findings to the organization The second essential step in the change process was gaining a broad base of acceptance. Here, the primary tasks were communication and persuasion, with the goal of obtaining organization-wide buy-in and visibility for the project and its findings. This step was especially important at Google because of its commitment to consensus-based decision making. Given employees’ deeply rooted aversion to top-down directives, the project’s findings were unlikely to be widely adopted if they were dictated or imposed by fiat; instead, employees had to be genuinely convinced of their value.¹⁶

To ensure acceptance, the Project Oxygen team (now broadened to include other members of people operations) engaged in extensive efforts to share and “socialize” their findings. They did so to first create a receptive environment for change.¹⁷ Among their key activities were securing support from senior executives; simultaneously pursuing bottom-up and peer buy-in by making a large number of presentations, each one tailored to a specific audience (e.g. engineers, sales, and G&A) but at the same time common in the willingness to share details about the research methodology; and

¹⁴ This approach is in keeping with standard statistical tests, in which “the null hypothesis...suggests that there is no statistically significant difference or relationship between a given set of observations.” See Thomas H. Davenport, and Jinho Kim, *Keeping Up with the Quants* (Boston: Harvard Business Review Press, 2012), p. 87.

¹⁵ Combining the views of superiors and subordinates (or peers) is important because the two groups often have different perspectives based on different information. A study of engineers at Bell Labs, for example, found that managers and engineers frequently disagreed about which engineers were the most productive. There was only a 50 percent overlap between the two groups; for this reason, researchers decided that “star performers” were only those who made it on to both managers’ and co-workers’ lists. See Robert E. Kelley, *How to Be a Star at Work* (New York: Times Business, 1998), esp. pp. 5-7, 269-270.

¹⁶ The need for “selling” to ensure the acceptance of new initiatives is hardly limited to Google. It is an essential ingredient in effective strategy implementation. See Donald C. Hambrick and Albert A. Cannella, “Strategy Implementation as Substance and Selling,” *Academy of Management Executive*, volume 3 (1989), pp. 278-285.

¹⁷ For more on the importance of creating a receptive environment for change, see David A. Garvin and Michael A. Roberto, “Change Through Persuasion,” *Harvard Business Review*, February 2005, pp. 104-112.

creating internal champions for the findings by spreading the word through tech advisors and other Googlers with high credibility among their peers.

At the same time, the team began the process of crystallizing and distilling the findings into a more condensed form. To that end, they updated the company’s Upward Feedback Survey (UFS) so that it was now based on the Oxygen 8 attributes, building upon a familiar, accepted instrument rather than introducing a completely new one. They created two separate versions of the survey instrument, one for the engineering organization and one for sales and G&A, customizing the tools so that the language and terminology were more tailored and accessible. They issued the surveys twice per year, but at different times than performance evaluations, clearly positioning the results as a developmental tool rather than as a potentially threatening performance metric. This positioning was not only consistent with Google’s developmentally-oriented culture; it also encouraged managers to approach the surveys with a learning rather than a performance mindset.¹⁸

At this stage of the change process, the project team had to meet several requirements for success. All involved breaking down entrenched views and packaging the research findings in accessible, actionable form. A broad-based communications campaign was essential, as was securing support from diverse constituencies.¹⁹ The goal was to convince employees that the proposed changes were not cumbersome or threatening, but enabling and supportive of their interests. Like the activities at the prior stage of the process, these involved another delicate balancing act: providing common, standardized approaches that at the same time were tailored to some degree to the distinctive practices of different departments and functional groups.

Improving management effectiveness at Google through widespread adoption of the Oxygen 8 attributes This step was aimed at formalizing the research findings, translating them into actionable behaviors, and embedding these behaviors in repeatable organizational processes and systems. The goal was to change the approach to managing at Google by creating a virtuous, reinforcing cycle of measurement, feedback, and learning.

To ensure that the desired management attributes were widely adopted, the Project Oxygen team developed and publicized granular guidelines, rules of thumb, and best practice examples. All were framed in actionable, behavioral terms. The team also developed a set of interactive training programs that were aimed at improving managers’ performance on particular attributes and facilitating the transition into managerial roles. They made these programs easily accessible, with simple sign-up processes (for example, a manager who received a low UFS score on a particular attribute would automatically receive an email invitation to sign up for a training course on that topic). Training courses were made available in multiple formats – intensive and drawn out, virtual and on-site – to accommodate diverse needs. To supplement these courses, the team arranged for presentations and panel discussions by functional experts who were recognized as highly effective managers. Engineering managers presented to other engineering managers, while sales managers spoke to other sales managers. This not only subtly shifted ownership and responsibility for institutionalizing the new behaviors from people operations to line managers; it also provided visible, credible role models. To reinforce this role modeling, the Project Oxygen team modified the criteria for the Great Manager Award so that they were closely aligned with the Oxygen 8 attributes.

¹⁸ For discussions of the distinction between the learning (growth) and performance (fixed) mindsets, see Carol S. Dweck and Ellen L. Leggett, “A Social-Cognitive Approach to Motivation and Personality,” *Psychological Review*, volume 95 (1988), pp. 256-273, and Carol S. Dweck, *Mindset* (New York: Random House, 2006).

¹⁹ For further discussion of the elements of an effective communications campaign, see John P. Kotter, *Leading Change* (Boston: Harvard Business School Press, 1996), ch. 6.

The team also positioned HR business partners as in-the-field supporters and sounding boards for managers needing help interpreting their UFS scores.

At this stage of the change process, the team had to meet several additional requirements for success. The primary challenges were building out the supporting infrastructure, routinizing the data collection, continuing to build acceptance and understanding of the Oxygen 8 attributes, and providing ongoing reinforcement of the desired behaviors and attitudes. In essence, the team’s goal now shifted to embedding the Oxygen 8 attributes in organizational systems and processes so that they became part of employees’ habitual routines.²⁰

Taken together, the three phases of Project Oxygen correspond closely to classical three-stage models of the change process.²¹ In these models, the first stage is a period of questioning, challenging, and consciousness-raising. Here, the goal is to unfreeze prevailing attitudes by generating deep and enduring dissatisfaction with the status quo, providing employees with the motivation to move forward with new ways of thinking and acting.²² It is at this stage that change leaders need to craft a vision or plan that explains where the organization is headed. The second stage is a period of changing, clarifying, and reinforcing. Both attitudes and behaviors need to be shifted, and employees must begin adopting new ways of working and interacting. The third stage is a period of consolidating, institutionalizing, and reviewing. Here, the goal is to make the desired changes stick, building them into the fabric of the organization so that they persist over time. Organizational structures, systems, processes, and norms often have to be altered so that they support new attitudes and behaviors. According to the preceding analysis, the three phases of Project Oxygen closely parallel these three stages of change, although the boundaries between stages are at times blurred and overlapping. In practice, the critical steps do not always occur in the simple, linear sequence described by change models.

The Oxygen 8 Attributes

Elements Eight management attributes constitute the core findings of Project Oxygen.²³ They represent a distillation of the PiLab’s research into the elements of effective management at Google and are aimed primarily at leaders of small groups and teams, especially first- and second-level managers. Collectively, they fall into two broad categories: managing people, and managing the work.

The attributes related to managing people have a largely developmental quality. They focus on expanding team members’ capabilities and skill sets, tailoring advice and interventions to individuals’ needs, and helping employees improve through feedback and coaching. The attributes in this category include: is a good coach, expresses interest/concern for team members’ success and well-being, helps with career development, and is a good communicator.

²⁰ For a discussion of the process of establishing new organizational routines, see Amy C. Edmondson, Richard M. Bohmer, and Gary P. Pisano, “Disrupted Routines: Team Learning and New Technology Implementation in Hospitals,” *Administrative Science Quarterly*, volume 46 (2001), pp. 685-716.

²¹ An example would be Kurt Lewin’s unfreezing, changing, and refreezing. See Lewin, *Field Theory in Social Science*.

²² Michael Beer, “Leading Change,” in John J. Gabarro, ed., *Managing People and Organizations* (Boston: Harvard Business School Press, 1992), pp. 424-426, and David A. Nadler, “The Effective Management of Organizational Change,” in Jay W. Lorsch, ed., *Handbook of Organizational Behavior* (Englewood Cliffs, NJ: Prentice-Hall, 1987), p. 364.

²³ For a model of managing that incorporates all of the Oxygen 8 attributes as well as a number of others, see Mintzberg, *Managing*, ch. 3.

The attributes related to managing the work embody a number of core principles of effective implementation: creating aligned goals, disseminating common, shared information, motivating team members, and eliminating roadblocks and barriers, especially those arising from outside the work group.²⁴ The attributes in this category include: empowers the team and does not micromanage, is productive and results-oriented, is a good communicator, and has a clear vision/strategy for the team.

Several attributes implicitly address the issue of balance, especially the tension between “doing” and “delegating.” Here, scholars argue that there is no single right approach, appropriate for all circumstances.²⁵ Instead, managers are best served by shifting their level of involvement depending upon the challenges facing their teams, the urgency of the problem, and the skills and capabilities of team members. At times, managers must be hands-on and actively involved in the work, rolling up their sleeves and pitching in to get the job done; at other times, they must delegate and take the “view from the balcony,” acting primarily as advisers, overseers, and monitors. The attributes in this category include: empowers the team and does not micromanage, and has key technical skills that help him/her advise the team.

Underlying Assumptions Together, the eight attributes provide a comprehensive (although incomplete) picture of the essential elements of effective management. It is important to recognize that both the attributes and associated recommendations for improving managers’ effectiveness rest on a number of unstated assumptions. Key assumptions include:

- Management consists of a set of describable, articulable behaviors.
- These behaviors are broadly applicable to diverse functional groups and differing nationalities and cultures, with only minor adjustments or adaptations.
- Managers can become more effective at these behaviors through a combination of feedback, reflection, training, practice, and coaching.
- Effective management is as much about developing people and ensuring employee commitment as it is about getting the work done.
- Effective management is more about raising employees’ level of accomplishment than it is about culling out “bad apples,” removing sub-par performers, or limiting dysfunctional behaviors.

These assumptions are, for the most part, supported by research. A number of scholars have studied managers *in situ* and have described, at a highly granular level, the actions, interactions, and behaviors they have observed. They have also noted the existence of common behaviors across diverse functions, positions, and cultures.²⁶ Studies of experiential learning have shown that people can learn and improve practical skills such as management through processes of feedback, reflection, and experimentation that closely match those used by Project Oxygen.²⁷ Organizational scholars have

²⁴ David A. Garvin, “Where Implementation Breaks Down: Why Can’t Companies Get the Job Done? *The Conference Board Review*, Summer 2013.

²⁵ Hill, *Becoming a Manager*, pp. 147-153, and Mintzberg, *Managing*, pp. 80-88.

²⁶ For two early summaries of this body of research, see Colin P. Hales, “What Do Managers Do? A Critical Review of the Evidence,” *Journal of Management Studies*, volume 23 (1986), pp. 88-115, and Leonard R. Sayles, *Leadership: Managing in Real Organizations*, Second Edition (New York: McGraw-Hill, 1989). For a more recent synthesis, see Mintzberg, *Managing*.

²⁷ See, for example, David A. Kolb, *Experiential Learning* (Englewood Cliffs, NJ: Prentice-Hall, 1984), and D. Scott DeRue and Ned Wellman, “Developing Leaders via Experience: The Role of Developmental Challenge, Learning Orientation, and Feedback Availability,” *Journal of Applied Psychology*, volume 94 (2009), pp. 859-875.

long argued that developing people and creating deep bonds with employees are among the most important responsibilities of managers at companies seeking high performance.²⁸ Only the last assumption may be somewhat peculiar to Google, since it presupposes the existence of a talented, highly motivated work force, which is not found at all companies.

Evaluation The eight attributes therefore rest on a firm foundation. They have a number of benefits to Google. Collectively, the attributes provide employees with a common language and a shared vocabulary for discussing the often amorphous topic of management. They also offer clear guidelines for improving performance. Widespread adoption of the eight attributes has led to quantifiable improvements in management effectiveness, raising managers’ percent favorable scores across all categories and geographies from 83% to 88%. Improvements have been most dramatic among the least effective managers, who are often the source of deep employee dissatisfaction. Moreover, management effectiveness, as measured by the eight attributes, has been shown to have a significant impact on how employees perceive life at Google, particularly their evaluations of the degree of collaboration, transparency of performance evaluation, and commitment to innovation and improvement.

Perhaps the greatest value of the eight attributes is that they create a straightforward, easy-to-follow, managerial checklist.²⁹ None of the attributes is especially novel or unexpected. Most seem obvious – a matter of common sense. The list is certainly not exhaustive or completely comprehensive.³⁰ Yet despite their simplicity and limited scope, the eight attributes serve a vital function: they remind managers of important things that they need to do on a regular, consistent basis in order to be effective. Many of these activities are likely to be overlooked in the press of day-to-day responsibilities. In many organizational settings, “The urgent drives out the important.” Firefighting and deadline-driven behavior are common preoccupations of managers.³¹ Longer-term needs, especially those involving people development, often get shortchanged in the process. Particularly for new managers, who can be easily overwhelmed by the job, a checklist provides a powerful tool for ensuring that essential but non-urgent tasks are not forgotten or underemphasized.

At the same time, because of their breadth the eight attributes might be viewed as little more than obvious, simplistic generalizations of limited value. This is especially true given the amount of high-powered research that went into developing them. Critics might accuse Project Oxygen of codifying

²⁸ Two representative examples are Charles O’Reilly and Jeffrey Pfeffer, *Hidden Value* (Boston: Harvard Business School Press, 2000) and Christopher A. Bartlett and Sumantra Ghoshal, “Building Competitive Advantage through People,” *MIT Sloan Management Review*, Winter 2002, pp. 34-41.

²⁹ For a discussion of the importance of using checklists to ensure consistent behavior in complex work environments such as aviation and medicine, see Atul Gawande, *The Checklist Manifesto* (New York: Picador, 2012). For a discussion of their use at GE to ensure a common, consistent approach to leading change, see David A. Garvin, *Learning in Action* (Boston: Harvard Business School Press, 2000), pp. 130-132.

³⁰ When it comes to checklists, comprehensiveness is not necessarily a virtue. Gawande notes, “Bad checklists are vague and imprecise. They are too long; they are hard to use; they are impractical...Good checklists, on the other hand, are precise. They are efficient, to the point, and easy to use even in the most difficult situations. They do not try to spell out everything...” See Gawande, *The Checklist Manifesto*, p. 120.

³¹ In her interviews with new managers, Hill observed, “Most described themselves as firefighters. They were too preoccupied with current problems to devote time or energy to thinking about potential problems...Most seemed to be quickly absorbing the managerial tendency to gravitate toward the more active aspects of their work – the ones that were current, specific, and well defined.” See Hill, *Becoming a Manager*, pp. 55, 63.

the obvious.³² The eight attributes might possibly lead to inauthentic behavior, with managers acting in ways that they believed led to higher “test scores” (i.e. UFS scores) even though they did not truly feel that the recommended activities were useful or desirable. Finally, the existence of semi-annual feedback ratings from their direct reports might, as Loux argues on page 10 of the case, lead managers to moderate or dilute their critiques of those same employees in order to avoid retaliatory responses.

What is one to make of these concerns? On the one hand, the eight attributes are indeed simple, unsurprising, and defined at an abstract, generic level. On the other hand, they are invariably accompanied by granular, highly concrete examples, with detailed descriptions of best practices and exemplary behaviors. As Exhibit 4 illustrates, this allows managers to understand both the overarching concept (“empowers the team & does not micromanage”) and the many different ways that it can be enacted (“balances giving freedom with being available for advice,” “advocates for me in the HR and compensation processes,” “sent me to one of our Europe offices – on my own! – to deliver the training”). The eight attributes thus allow for considerable individual variation and tailoring; they are less bland generalizations or rigid formulas than inclusive guidelines. Moreover, because they focus on behaviors rather than attitudes the concerns about inauthenticity are likely to be overblown. It is difficult to “fake” the attributes in pursuit of higher test scores. Either managers have demonstrated the indicated behaviors or they have not. The UFS does not ask employees to assess their managers’ motivations, values, or beliefs; instead, it asks them to evaluate the extent to which their managers have demonstrated a particular set of behaviors. One implication of this focus on behaviors is that despite Loux’s concerns, giving an employee a low performance rating had only a miniscule (less than 1%) impact on the UFS scores that a manager received from the employee.

Project Oxygen does, however, prompt two questions. First, do the eight attributes provide Google with a source of sustainable competitive advantage? Put another way, does a commitment to managerial excellence lead to superior long-term performance?³³ In terms of sustainability, one potential concern is “evaluation overload.” The UFS depends on the goodwill of employees; they voluntarily agree to respond on a semi-annual basis. However, the UFS is only one among many surveys that employees are asked to complete. What if Googlers decided that they were spending too much time filling out surveys? Would response rates fall to the point where the results were no longer representative or valid? Sustainability also depends upon the continued effectiveness of managers who are proficient at the eight attributes, as well as the relevance of those attributes to senior executive positions. The case notes that in the most recent round of promotions to vice president, a disproportionate number of those promoted were winners of the Great Manager Award, which is based heavily on the eight attributes. Will those same attributes apply to senior positions, where leadership skills may be equally or even more important than management skills?³⁴

Whether management excellence provides Google with a long-term edge in performance is also open to question. While UFS scores are clearly related to a wide range of measures of employee satisfaction and perceptions of the work environment, it is unclear whether they also translate into

³² For an article that takes this perspective, see Adam Bryant, “The Quest to Build a Better Boss,” *New York Times*, March 13, 2011, Sunday Business, pp. 1, 7.

³³ These questions are closely related to the question of whether culture can be a source of sustainable competitive advantage. Jay Barney argues that for that to happen, three conditions must be met: the culture must be valuable, rare, and imperfectly imitable. See Jay B. Barney, “Organizational Culture: Can It Be a Source of Sustained Competitive Advantage?” *Academy of Management Review*, volume 11 (1996), pp. 656-665.

³⁴ There is now a large literature on the differences between management and leadership. For an introduction to the topic, see John P. Kotter, *A Force for Change* (New York: Free Press, 1990).

such bottom-line measures as increased sales, improved productivity, and greater innovation. These relationships are difficult to establish definitely because there are so many possible causal and confounding factors.³⁵ The eight attributes and the associated surveys and training tools also appear to be imitable and replicable, since there is little proprietary about the Project Oxygen approach. This suggests that if performance benefits exist, they are likely to be transitory since other firms will eventually copy Google’s practices.³⁶ Taken together, these arguments suggest that Google’s commitment to the Oxygen 8 attributes, while offering tangible short-term benefits, may not provide a sustainable long-term performance advantage. However, a definitive answer requires empirical evidence that is currently lacking.

The second question prompted by Project Oxygen is whether the eight attributes are peculiar to Google, or whether they would apply equally well to other organizations. On the one hand, Google clearly has a number of distinctive characteristics: a uniquely smart, skilled, and motivated work force; a heavy reliance on human and knowledge capital rather than physical assets; a deeply ingrained developmental culture; and abundant slack resources that can be used for HR research, training programs, and other efforts aimed at making the company a more desirable place to work. On the other hand, the eight attributes have many characteristics that are generic and generalizable. They are broadly consistent with academic research on managerial behavior and effectiveness, employee commitment, and the distinctive needs of knowledge workers, suggesting that they are not wholly limited to Google but are likely to apply, with minor variations, to firms in similar settings (e.g. Silicon Valley), having similar competitive profiles (e.g. high technology firms, especially those based on software), with work forces comprised largely of skilled knowledge workers (e.g. consulting firms).³⁷ The ranking and relative importance of the attributes may differ, but the basic findings are likely to apply to many firms beyond Google.

Setty’s Agenda: Future Projects

Setty is considering four projects, all of them extensions of Project Oxygen. They include: (1) expanding the analysis to the most senior managers at Google, such as directors and vice presidents, (2) focusing on the life cycle of managers, including hiring, lateral job transitions, promotions, and departures, (3) focusing on the composition and functioning of teams, and (4) trying to create a larger number of “truly amazing managers.” Because the people analytics group is small, with limited time and resources, Setty must choose among these projects. To do so, he needs some criteria for making the decision. Two useful criteria are how much value-added the project is likely to provide Google, and how well the project lends itself to Project Oxygen’s proven methodology and approach.

Using these criteria, the first option, expanding the analysis to the most senior managers at Google, has the following advantages and disadvantages. On the plus side, it could have potentially

³⁵ In a series of studies, Nicholas Bloom and his colleagues have shown that superior management practices that involve performance monitoring, target setting, and incentives/people management are related to greater profitability, increased productivity, faster growth, and longer survival periods for firms around the world. However, these practices are far more operational and policy-focused than the Oxygen 8 attributes. See Nicholas Bloom et al., “Management Practices Across Firms and Countries,” *Academy of Management Perspectives*, February 2012, pp. 12-33.

³⁶ Michael Porter has made a similar point about the limitations of operational excellence as a source of strategic advantage. He argues, “Constant improvement in operational effectiveness is necessary to achieve superior profitability. However, it is not usually sufficient. Few companies have competed successfully on the basis of operational excellence over an extended period...The most obvious reason for that is the rapid diffusion of best practices. Companies can quickly imitate management techniques...” See Michael E. Porter, “What Is Strategy?” *Harvard Business Review*, November-December 1996, p. 63.

³⁷ For advice on how best to manage knowledge workers, see Thomas H. Davenport, *Thinking for a Living* (Boston: Harvard Business School Press, 2005).

large performance impacts because senior managers have such significant responsibilities. Senior managers establish corporate policies, determine strategic direction, and allocate resources, all activities with clear bottom-line impacts. This option would also be relatively easy to execute because the Project Oxygen methodology could be applied with few changes. On the minus side, because Google is such a flat organization and has a relatively small group of senior managers, differences in performance and effectiveness may be negligible at senior levels, making it difficult to generate enough variance in the data to produce meaningful statistical results. Measurement may also be complicated by the fact that successful senior executives may well be distinguished by subjective, difficult-to-assess attributes such as vision or the ability to inspire others. Most senior executives are “managers of managers;” this role is likely to require a different skill set than that displayed by lower-level managers, who are much closer to those who are directly involved in design, production, and sales activities.³⁸ It also complicates the data collection process, since the effectiveness of senior executives at such tasks as setting and communicating strategic direction probably needs to be assessed by employees at multiple organizational levels rather than by direct reports alone.

The second option, focusing on the life cycle of managers, has the following advantages and disadvantages. On the plus side, it is the most straightforward extension of Project Oxygen and would therefore be easiest to implement. Several elements of the life cycle model (e.g. the impact of training on new managers’ UFS scores) are already in place. In many ways, this option represents the completion of Project Oxygen; by adding a dynamic element – the movement of people into and out of managerial roles through hiring, lateral transfers, and demotions – it would provide a fuller picture of how, when, and why managers are more or less effective in their assignments and what personal characteristics or experiences help determine their success. Here too, however, the necessary data may be difficult to collect, especially if it involves managers’ performance at jobs prior to joining Google. Moreover, given the company’s relatively low turnover rates and the likely small number of persistently weak managers, the resulting findings may provide only limited value-added.

The third option, focusing on the composition and functioning of teams, has the following advantages and disadvantages. On the plus side, the potential contribution is enormous, since so much of Google’s work is done in teams. Analysts would be able to build on an extensive, well-regarded academic literature, with widely accepted measures of team effectiveness, team design, and team behaviors.³⁹ This option would also extend Project Oxygen from its largely individualistic focus (UFS scores are constructed from the evaluations that individual employees make of individual managers, based on behaviors that they have personally observed or experienced) to a more collective, group focus. On the minus side, the number of variables could quickly become unmanageable, given the large number of interaction effects when multiple people are involved. Particularly when it comes to evaluating the precise mix of attributes, personalities, and experiences that lead to superior team performance, statistical analysis is likely to be difficult, and meaningful results may not materialize.

The fourth option, trying to create a large number of truly amazing managers, has the following advantages and disadvantages. On the plus side, this option would involve a straightforward application of the Project Oxygen methodology to a limited sub-group of managers such as winners

³⁸ See Ram Charan, Stephen Drotter, and James Noel, *The Leadership Pipeline* (San Francisco: Jossey-Bass, 2001) for a discussion of the roles and responsibilities of managers at different levels.

³⁹ For a summary and synthesis of over 50 years of research on team effectiveness, see Steve J. Kozlowski and Daniel R. Ilgen, “Enhancing the Effectiveness of Work Groups and Teams,” *Psychological Science in the Public Interest*, volume 7 (2006), pp. 77-124. For a detailed discussion of team design, see J. Richard Hackman, *Leading Teams* (Boston: Harvard Business School Press, 2002). For an in-depth discussion of team behaviors, see Amy C. Edmondson, *Teaming* (San Francisco: Jossey-Bass, 2012).

of the Great Manager Award. Because of their small numbers, this group could be studied in-depth through ethnographic analysis. The findings would likely build upon the Oxygen 8 attributes, since the criteria for winning the Great Manager Award are already based on those attributes. On the minus side, it is not clear what Setty means by the phrase “truly amazing managers,” or whether there is agreement within Google about who qualifies for that title. Nor is it clear that truly amazing managers produce truly amazing bottom-line results. The links to performance are not yet clearly established; in fact, Loux argues on page 10 of the case that managers with extremely high UFS scores may be muting their criticisms in hopes of higher ratings, “avoiding problems by not having tough conversations with their direct reports.” Finally, there is the question of whether greatness can be taught. Becoming a truly amazing manager may be more a reflection of temperament, personality traits, and core values than a matter of concrete, learnable behaviors.

All of the proposed projects therefore have strengths and weaknesses. Based on the above analysis, a case can be made for any one of the options. The third and the fourth options, however, despite the difficulties in execution, seem likely to produce the greatest impact and draw most effectively on the research methods already pioneered by Project Oxygen.

Discussion Questions

Culture and People

1. How would you describe Google’s culture?
2. What are the key features of an engineering-dominated culture?
3. Both Page and Setty describe Google as “a family.” What are they saying about the organizational culture?
4. Others describe the culture as “consensus-oriented.” What kinds of behavior does that suggest?
5. What was the view of managers prior to Project Oxygen?
 - Why did engineers view managers so negatively?
 - What was their (often unstated) view of the manager’s job/role?
 - What do you think it was like being a low- or mid-level manager in Google’s early years?
6. What kinds of people work at Google? What are their personal qualities?
7. What should be the role of managers in such a setting? The role of the HR function?
8. What do Google’s employees want from their managers? What do they need from their managers?

Project Oxygen as a Successful Change Process

1. What is your evaluation of the steps the Project Oxygen team took to identify the characteristics of effective managers at Google? What were the biggest challenges at this stage of the change process? How were these challenges overcome?

2. What is your evaluation of the steps that the team took to roll out Project Oxygen’s findings to the organization? What were the biggest challenges at this stage of the change process? How were these challenges overcome?
3. What is your evaluation of the steps the team took to improve management effectiveness through widespread adoption of the Oxygen 8 attributes? What were the biggest challenges at this stage of the change process? How were these challenges overcome?

The Oxygen 8 Attributes

1. What is your assessment of the Oxygen 8 attributes?
 - What value (if any) do they provide?
 - Are they just broad generalizations, similar to what one would find in an introductory management textbook?
 - Is the approach formulaic? Is it a form of social engineering?
 - Are the attributes likely to create inauthentic managers, who adopt behaviors primarily to get higher “test (UFS) scores?”
 - What (if anything) surprised you about the list of attributes?
2. What assumptions underlie the research findings and the associated recommendations for improving management effectiveness?
3. Do the attributes provide Google with a sustainable source of competitive advantage?
4. Are the findings generalizable, or do they apply primarily to Google? Would they work in:
 - A government agency?
 - A nonprofit organization?
 - A large corporate bureaucracy?
 - An organization with unmotivated, uncommitted employees, who were there just for the paycheck?
 - Non-U.S. cultures and countries?
5. Think of the very best manager you ever had. How would you evaluate him or her on the Oxygen 8 attributes? What were his or her primary strengths and weaknesses?
6. Think of the very worst manager you ever had. How would you evaluate him or her on the Oxygen 8 attributes? What were his or her primary strengths and weaknesses?
7. Try to apply the list of attributes to your own performance as a manager. Where do you feel subordinates would rate you highly? Where would they give you a low score? What does this tell you about needed areas of improvement?

Setty’s Agenda: Future Projects

1. What should be Setty’s priorities going forward? Which projects should he pursue? Why?
2. What criteria should he use for choosing among projects?
 - Which projects best lend themselves to the Project Oxygen methodology and approach?
 - Which projects are likely to provide the greatest value-added to Google?

3. Let’s focus for the moment on the goal of creating “truly amazing managers.”
 - Have any of you had a truly amazing manager? What was it like?
 - What did he or she do that was distinctive?
 - What approach should the PiLab take to studying the characteristics and development of truly amazing managers?
 - Do the Oxygen 8 attributes already capture the critical aspects of truly amazing managers? Or does something else make them special?

Teaching Suggestions

Culture and People The instructor can begin with an open-ended question – “How would you describe Google’s culture?” – to get the discussion started. Students will quickly generate a list of key characteristics. Here, it is important to push students to a deeper analysis of what it means to be engineering-dominated. The instructor can say, “On page 2 of the case, Eric Flatt observes that Google is ‘a company built by engineers for engineers.’ For those of you who are engineers or have worked at companies with a similar ethos, what are the distinguishing features of such a culture? What do engineers value? Why?” Eventually, a student is likely to note that engineers do *not* value managers, seeing them as impediments and barriers rather than enablers. This is an important point, and the instructor should probe on why this view is so pervasive and deeply held, particularly at Google in its early years.

Once this discussion has run its course, the instructor should shift to the topic of the kinds of people who work at Google. He or she can say, “We’ve now established the basics of Google’s culture. Let’s take a closer look at the people who work there. What are they like?” Again, students will quickly generate a list of key characteristics. Here, it is important to deepen the analysis by pushing students to understand the implications of having a company filled with young, driven, over-achievers. The instructor should ask, “What do people with these characteristics need or want from their managers?” It is important to ask the question with this wording because students will soon observe that what employees *want* (autonomy and freedom to get the job done their way, as well as lots of positive reinforcement) often conflicts with what they *need* (guidance and direction, as well as critical oversight and feedback). Follow-up probes include: “What should be the role of managers in such a setting? What should be the role of the HR function?” In total, the discussion of culture and people lasts 20-25 minutes.

Project Oxygen as a Successful Change Process To transition to this topic, the instructor can say, “We’ve now set the stage for Project Oxygen. The entire project can be viewed as a change process designed to improve management effectiveness. Members of the PiLab had to navigate the twin challenges that we have just discussed – a lack of respect for managers and a conflict between what employees at Google say they want from managers and what they actually need. The resulting change process unfolded in three broad stages: identifying the characteristics and impact of effective managers, rolling out the findings to the organization, and improving management effectiveness through widespread adoption of the Oxygen 8 attributes. Let’s take them one step at a time.” Students should discuss each of the stages in turn, listing key steps and activities. Here, the instructor’s role is to follow up on critical items that students list by asking why the PiLab team proceeded as it did, what challenges had to be overcome, and why the approach was successful. Examples of useful probes include: “Why was it important that the team started with the hypothesis that managers *don’t* matter? Why did the team spend so much time making presentations of its findings, including detailed descriptions of the research methodology? Why did the team insist on taking a developmental approach to the eight attributes rather than incorporating them in

performance evaluations?” After the key steps have been examined, the instructor can wrap up this segment of class by noting that the three stages of Project Oxygen mirror those described in classical models of planned change processes such as Lewin’s unfreezing, changing, and refreezing. In total, this discussion lasts about 15-20 minutes.

The Oxygen 8 Attributes The instructor can introduce this part of the discussion by asking another broad, open-ended question: “What is your assessment of the Oxygen 8 attributes?” Students normally divide into two camps. One group will see the attributes as valuable, primarily as a way of ensuring that important but mundane management tasks are not overlooked. They will applaud the granularity of the findings and the best practice examples, particularly as a set of behavioral guidelines for educating new managers. Another group will take the opposite view, arguing that the eight attributes are obvious and simplistic. They will criticize Project Oxygen as high-powered research that has produced little of value – findings that could easily have been gleaned from an introductory management textbook. The instructor should allow this debate to percolate for a while, with limited intervention. At some point, however, the instructor should ask: “If these attributes are so obvious, why is good management so hard to do?” This question highlights the value of the attributes as a managerial checklist and leads to a discussion of the many competing demands on managers’ time and attention. If time permits, the instructor can ask follow-up questions about the assumptions about management that lie behind Project Oxygen, whether the eight attributes are likely to lead to formulaic, inauthentic responses from managers, and whether they believe that the findings are generalizable to other companies and industries or are peculiar to Google.

To enrich and personalize the discussion, the instructor can have students conduct a simple exercise: evaluating the best and worst managers they have had, using the Oxygen 8 attributes listed in Exhibit 5 of the case. Students should be encouraged to “share and compare,” talking with those seated next to them in order to determine if their experiences had anything in common. After a few minutes of small-group discussion, the instructor should say, “Let’s compare notes. What were some of the characteristics of the best managers that you have had? What were some of the characteristics of the worst managers? What conclusions do you draw about the challenges of being an effective manager?” This portion of class lasts 20-40 minutes, depending on the amount of time that the instructor chooses to devote to the exercise.

Setty’s Agenda: Future Projects The instructor can shift to this topic by saying, “The findings of Project Oxygen have now been widely adopted, and Setty is considering several new projects for the people analytics group. What should be his priorities going forward? Why?” Students will argue for all four options; to structure the discussion, the instructor should create two columns on the blackboard, Options and Criteria, and sort comments into each one. Eventually, the instructor should ask, “Let’s step back from the four options and dig deeper into the basis for making this decision. We now have a long list of criteria that were cited in support of the different options. Which ones should Setty use for making the choice? Why?” If time permits, the instructor can probe further on the topic of truly amazing managers, asking students if any of them have had such a manager, what their experiences were like, whether Google should try to increase their numbers and, if so, what approach it should take. In total, this discussion lasts 10-15 minutes.

Conclusion The instructor can conclude class in two different ways. One is to use Project Oxygen as an exemplar of an effective planned change process. In particular, the project neatly

illustrates the “change formula” developed by Michael Beer.⁴⁰ Beer argues that the amount of organizational change can be represented as follows:

$$\text{Amount of Change} = (\text{Dissatisfaction} \times \text{Model} \times \text{Process}) > \text{Cost of Change}$$

There are four variables in this framework. “Dissatisfaction” is the shared level of dissatisfaction with the status quo. “Model” is a description or picture of the desired future. “Process” is the sequence of activities and events aimed at gaining buy in and acceptance. “Costs of change” are the losses of power, competence, identity, and rewards that employees anticipate from the proposed changes. Beer argues that for change to succeed, the first three variables must, in combination, be large enough to overcome resistance from employees due to the perceived costs of change. As an earlier discussion has shown, Project Oxygen targeted all four variables in the change formula. It increased dissatisfaction with the status quo by providing evidence against the prevailing view of managers as non-value-adding, developed a model for the future by describing eight essential management attributes, rolled out the findings slowly over time in order to gain buy in and acceptance, and minimized the perceived losses of employees by ensuring that the process was not linked to performance evaluations but was positioned as a developmental effort accompanied by extensive training and support.

A second approach to concluding class is to focus on the lessons for effective management practice. These include the importance of clearly specifying desired behaviors, the need to coach and develop subordinates, the value of taking a developmental perspective, the power of having a short, actionable checklist of essential activities, and the virtues of measurement and analysis in identifying and communicating the value-added of managers.

See **TN Exhibit 1** for a sample blackboard plan.

⁴⁰ Beer, “Leading Change,” pp. 424-431.

TN Exhibit 1 Sample Blackboard Plan

